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Research Interests

Research Question: Reality is abstract and high-dimensional. However, what we perceive are but fragments of it, colored by our own minds. From these fragments, how might we come to know the world more deeply?

Keywords: Optimization for AI; Statistical Estimation.

Education

Seoul National University Mar 2020 – expected Aug 2027

B.Sc. in Mathematics GPA: 4.02/4.3
B.Sc. in Mathematical Modeling for Complex Systems GPA: 4.3/4.3
B.Eng. in Architectural Engineering GPA: 3.96/4.3

Completed mandatory military service.

Purdue University Aug 2025 – Dec 2025

Exchange student in Mathematics and Industrial Engineering. GPA: 4.25/4.3

Manuscripts in Preparation

AAAAAAA (TBD) Sep 2026 – Dec 2026 (Expected)

In preparation for submission to Mathematical Programming.

Inverse Variational Inequalities: Identification and Statistical Estimation Feb 2026 – Nov 2026 (Expected)

In preparation with Taewoo Lee for submission to Operations Research (INFORMS).

Statistical estimation theory for inverse variational inequalities; parameter identification and statistical interpretation of ReLU equilibrium layers and monotone deep equilibrium models (monDEQs).

Research Experience

Optimization and Algorithms Lab, SNU Math Sep 2026 – present

Advisor: Dabeen Lee

- Studying looped Transformers and fixed-point reasoners.
- Co-authoring a manuscript.

Operations Management, SNU Business

Advisor: Taewoo Lee

Research collaboration Feb 2026 – present

Statistical estimation for inverse variational inequalities, with applications to ReLU equilibrium layers and monotone deep equilibrium models.

Reading and study Apr 2025 – Aug 2025

- Studied linear programming (Bertsimas) and convex optimization (Boyd).
- Read classic optimization papers from *Operations Research* and *Mathematical Programming*.

Awards and Scholarships

Dukshin EPC Scholarship for Academic Excellence Fall 2026

Dean's List and Semester Honors, Purdue University Fall 2025

Global Leadership Program Exchange Scholarship Fall 2025

SNU College of Engineering; support for study abroad.

SNU DAAE Alumni Scholarship for Academic Excellence Fall 2023

Sole recipient, Department of Architecture and Architectural Engineering.

Selected Projects

Perturbation Analysis of a Pre-trained SPO Framework Nov 2025 – Dec 2025

Investigating graph-model pre-training with conventional losses followed by SPO+ fine-tuning, and studying perturbation robustness through distributionally robust optimization and assumptions on graph topology.

Operating Cost Optimization for Logistics Centers 2024 Architecture Graduation Exhibition

Simulated demand in Seoul; used Gurobi and OSMnx to solve vehicle-routing and traveling-salesperson problems and visualize routes. Optimized photovoltaic installation area to reduce building energy costs.

Service and Leadership

Society of Engineering Network and Service Sep 2023 – Feb 2025

Led engineering outreach for high-school students and STEM activities for middle-school students; participated in educational programs for underrepresented students.

Student Council Mar 2021 – Dec 2021

Managed digital communications and designed event announcements, campus banners, and graduation-exhibition posters.

Skills

Programming and tools: Python, MATLAB, R, Julia, C/C++, LaTeX, Gurobi, OSMnx.

Engineering software: Midas, ETABS, Rhino, Grasshopper, AutoCAD.

Languages: Korean (native), English (advanced).

Selected Coursework

Seoul National University

Course	Term	Grade
Mathematical Algorithms 2 *(Graduate; Transformers)*	Fall 2026	In progress
Topics in Mathematics 2 *(Analytic Number Theory)*	Fall 2026	In progress
Algorithm	Summer 2026	A+
Measure Theory and Probability	Spring 2026	A−
Introduction to Mathematical Analysis with Practice 1	Spring 2026	A+
Differential Equations	Summer 2025	A+
Advanced Calculus 1	Spring 2025	A+
Linear Algebra	Spring 2025	A+
AI & Big Data Basics	Spring 2025	A0
Microeconomics	Summer 2023	A0
Macroeconomics	Summer 2021	A+

Purdue University — 2025 Fall

Exchange

Course	Code	Grade
Introduction to Convex Optimization *(Graduate)*	IE56100	A+
Linear Programming & Optimization Techniques	MA42100	A0
Operations Research – Stochastic Models	IE33600	A+
Probability	MA41600	A+
Introduction to Discrete Mathematics	MA37500	A+
Engineering Economics	IE34300	A+